

MethaneSAT™

Oil and Gas Infrastructure Mapping (OGIM) Database v2.7

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Overview

The Oil and Gas Infrastructure Mapping (OGIM) database is developed by Environmental Defense Fund (www.edf.org) in support of the MethaneSAT satellite project managed by MethaneSAT, LLC (www.methanesat.org). MethaneSAT, LLC is a wholly-owned subsidiary of Environmental Defense Fund (EDF). The primary objective of developing a global, granular oil and gas (O&G) infrastructure database is to support MethaneSAT's emission quantification, source characterization, and subsequent scientific- or advocacy-relevant analyses of methane emissions from the global oil and gas sector. This database can be used to support other related research works on global O&G methane emission assessment and source characterization.

This document describes a version of the OGIM (v2.7) based solely upon publicly available source datasets. The spatial coverage of OGIM v2.7 includes six continents and 152 countries. Oil and gas infrastructure records were obtained via a combination of manual and semi-automated search of O&G datasets in the public domain. We compiled, cleaned, and integrated 188 O&G geospatial datasets, which were combined within one standard data schema and coordinate reference system.

An in-depth description of the methods for the global OGIM database development, and a detailed discussion of the key applications of this database, can be found in the below-mentioned Earth System Science Data article. Note that the content of the v2.7 database accompanying this README is an updated version of the OGIM GeoPackage available on ESSD's website. Please cite this article when using the database in presentations and/or any publications:

Omara, M., Gautam, R., O'Brien, M. A., Himmelberger, A., Franco, A., Meisenhelder, K., Hauser, G., Lyon, D. R., Chulakadabba, A., Miller, C. C., Franklin, J., Wofsy, S. C., and Hamburg, S. P.: Developing a spatially explicit global oil and gas infrastructure database for characterizing methane emission sources at high resolution, *Earth Syst. Sci. Data*, 15, 3761–3790, <https://doi.org/10.5194/essd-15-3761-2023>, 2023.

Changes since v2.5.1

Version 2.5.1 aimed to incorporate the latest published O&G data globally as of January 2024, with North American datasets updated as of April 2024. For v2.7, the authors revisited every one of OGIM's data sources and incorporated the latest information available as of February 2025. About 42% of all well records in v2.7 were published by their original source between 1 February and 10 February 2025. About 23% of all pipeline records in v2.7 were published between 1 February and 10 February 2025, as well. Taking all 14 remaining infrastructure categories together, about 44% of the records in these categories were published between 1 February and 10 February 2025.

A number of new data sets were added to v2.7, including:

- Nonvertical wells in Saskatchewan
- Argentina pumping stations
- Nebraska wells
- South Dakota wells
- United States above ground LNG storage facilities
- State-specific wells in Western Australia, South Australia, and Queensland, to complement offshore and national wells reported by Australia's National Offshore Petroleum Titles Administrator (NOPTA)

For some regions and infrastructure categories, the authors replaced the underlying data source with a more frequently updated and/or more attribute-rich one. These include:

- Tennessee wells
- Missouri wells
- Oklahoma wells
- Colombia wells
- Germany wells (specifically Lower Saxony)
- Netherlands O&G fields and offshore platforms
- United Kingdom offshore platforms and petroleum terminals
- United States offshore platforms

Other Updates

- **Bug fix:** Turkey refineries were erroneously excluded in v2.5.1; the six refineries are now present.
- **Data availability:** Includes VIIRS flaring detections from the year 2023.
- **Quality assurance:** Additional duplicate records were removed.

Known Issues

- Various records in Argentina contain improperly encoded characters. Our team has determined this is due to corrupted information in the original dataset, and we are still working on resolving the issue.

Database structure

OGIM v2.7 is a collection of data tables within a [GeoPackage](#), an open-source geospatial database format. Each data table or “layer” of the GeoPackage, listed in **Table 1**, represents an infrastructure category that plays a major role in the O&G sector. All records in the database have an associated spatial point location, except for O&G pipelines (geometry type: *LineString*) and O&G fields, license blocks, and basins (geometry type: *Polygon*). These data have been transformed to a common spatial reference system (WGS 1984, EPSG:4326). All GeoPackage layer names use underscores to separate words.

Table 1: List of the geospatial data layers (representing O&G infrastructure categories) in the OGIM database.

Geopackage layer name	# of variables	# of records
Crude_Oil_Refineries	22	692
Equipment_and_Components	19	98,047
Gathering_and_Processing	26	10,396
Injection_and_Disposal	24	14,209
LNG_Facilities	24	547
Natural_Gas_Compressor_Stations	26	12,156
Natural_Gas_Flaring_Detections	22	10,233
Offshore_Platforms	19	3,903
Oil_Natural_Gas_Pipelines	24	1,858,109
Oil_and_Natural_Gas_Basins	13	709
Oil_and_Natural_Gas_Fields	13	17,742
Oil_and_Natural_Gas_License_Blocks	13	2,833
Oil_and_Natural_Gas_Wells	20	4,537,369
Petroleum_Terminals	24	3,661
Stations_Other	19	8,468
Tank_Battery	24	132,220

In addition to the 16 layers in the table above, the GeoPackage includes a table layer (no geometry information) called “Data_Catalog.” Each unique publication or web portal used as a source during OGIM database development is assigned a numeric Source Reference ID, or ‘SRC_ID’, and is listed in the Data Catalog alongside some metadata information. Each record in the final OGIM database has a ‘SRC_REF_ID’ attribute that can be used to join the record to its original source information in the Data Catalog, making it easy for a user to visit the original source(s) of the record if desired.

All geospatial data were pre-processed, analyzed, assembled, and tested using open-source software, including but not limited to Python v3.7 and QGIS v3.24. OGIM v2.7 uses UTF-8 encoding.

Data attributes

A full description of the database schema used in all layers can be found in **Supplement 1: OGIM v2.7 Database Schema**. A full description of the schema used in the “Data Catalog” layer can be found in **Supplement 2: OGIM v2.7 Data Catalog Schema**. Both supplements are provided with this README.

Some additional notes on data attributes:

- **Unique IDs:** Each record in each infrastructure layer is assigned a unique numerical identifier ('OGIM_ID'). The IDs do not repeat between data layers. In addition, each record has an assigned 'SRC_REF_ID', which identifies the original data source in a separate “Data Catalog” layer.
- **Missing values** (i.e., values missing or not reported in the original source) are handled by assigning “N/A” to string attributes, -999 to numerical attributes, and a generic date of “1900-01-01” to date/datetime attributes.
- **For date attributes** (i.e., 'SRC_DATE', 'INSTALL_DATE', 'SPUD_DATE', 'COMP_DATE'), if an incomplete date is recorded in the original data source, we fill out the rest of the date with '01'. For example, if a data source indicates a well was spudded in May 2019, with no day of the month provided, we record this date as “2019-05-01”. If a source indicates a facility was installed in the year 2017, we record this date as “2017-01-01.”
- **Operator names** have not been altered in any way from the original source of data and are assumed to be accurate as of the original source’s publication date.
- **Non-English source data:** Some values from non-English datasets (such as status, facility type, and drilling direction) were translated to English before inclusion in OGIM. However, place names, facility names, and operator names were kept in their original language.

Table 2: Sample record from OGIM's Oil and Natural Gas Wells layer.

Attribute name	Value
OGIM_ID	2081355
CATEGORY	OIL AND NATURAL GAS WELLS
REGION	CENTRAL AND SOUTH AMERICA
COUNTRY	ARGENTINA
STATE_PROV	SANTA CRUZ
SRC_REF_ID	105
SRC_DATE	2025-02-04
ON_OFFSHORE	ONSHORE
FAC_NAME	YPF.SC.EG-556
FAC_ID	97315
FAC_TYPE	OIL
DRILL_TYPE	CONVENTIONAL
SPUD_DATE	2003-02-15
COMP_DATE	2003-03-10
FAC_STATUS	ACTIVE
OGIM_STATUS	PRODUCING
OPERATOR	YPF S.A.
LATITUDE	-46.33622
LONGITUDE	-69.19831
geometry	POINT (-69.19831 -46.33622)

Descriptions of infrastructure categories in v2.7

Oil and Natural Gas Wells: contains point locations of O&G wellheads. Wellheads are located on well sites and each well site may have one or multiple wellheads. Wellheads may produce oil, gas, or both. The facility status attribute indicates the operating status of each wellhead. Other well types that assist with O&G operations (e.g., water injection, salt injection) are included in this layer, and are indicated by their FAC_TYPE attribute where available. Since this well dataset is intended for methane emission analysis and visualization of O&G infrastructure locations, we did not include locations that states reported as stratigraphic test wells. We also removed locations of wells that were proposed but never materialized (cancelled permits that were never drilled, for example).

Natural Gas Compressor Stations: natural gas compressor stations regulate the flow and pressure of natural gas for pipeline transportation of natural gas (e.g., within and across states). Often, but not always, these stations are located at points along a transportation pipeline.

Gathering and Processing: facilities designed to “gather” and treat or process O&G products, for example, to produce pipeline-quality natural gas, or remove impurities and other entrained

hydrocarbons in the oil and gas stream. Examples include natural gas processing plants, central gathering facilities, and crude oil treatment facilities.

Tank Battery: a system or group of storage tanks receiving oil and gas produced from one or multiple wellheads. May be located at a well site or at a major O&G gathering/processing facility (e.g., central gathering facility, compressor station and processing plant).

Please note that many states and provinces do not publish locations of tank batteries; for this layer in particular, lack of tanks in O&G producing regions most likely implies a gap in reporting rather than the absence of infrastructure.

Offshore Platforms: offshore production, gathering, and processing facilities, which could include an offshore drilling platform (a rig positioned above an offshore well), or a central processing platform (where hydrocarbons are processed or refined offshore before transit elsewhere). Offshore platforms may produce oil, gas, or both.

LNG Facilities: export and import liquified natural gas (LNG) facilities that treat and condense natural gas into a liquid form to facilitate the material's transport or sale. Often, but not always, these facilities are found near bodies of water and handle LNG transported by sea.

Crude Oil Refineries: crude oil refineries convert crude oil and other hydrocarbons into useful petroleum-based products, including, for example, production of transportation fuels, such as diesel and gasoline.

Petroleum Terminals: a facility that stores crude oil and refined petroleum products. Often, but not always, found co-located with a marine port or at the start of a pipeline system. Can be collocated with refineries, LNG facilities, and other major O&G facilities where O&G processing occurs.

Injection and Disposal: facilities used to safely dispose O&G waste, for example, via injection into deep, confined rock formations.

Equipment and Components: locations of O&G equipment and components at larger facilities (e.g., valves and dehydrators).

Stations – Other: includes some “minor” O&G facility types, such as metering and regulating stations, POL (petroleum, oil, and lubricants) pumping stations, and LACT (lease automatic custody transfer units). These types of facilities may be collocated with other major O&G gathering and processing facilities, e.g., compressor stations and processing plants.

Natural Gas Flaring Detections: locations of a facility or cluster of facilities where natural gas flaring has been detected based on [observations from the Visible Infrared Imaging Radiometer Suite \(VIIRS\)](#) satellite instrument. Gas flaring can occur at a wide variety of oil and gas facilities, including well sites, offshore platforms, compressor stations, processing plants, refineries and LNG facilities. The layer includes flaring detections and estimated flaring volumes for 2023. The Earth Observation Group at the Colorado School of Mines has granted EDF permission to include these 2023 data in OGIM.

Oil and Natural Gas Pipelines: pipelines that transport crude oil or natural gas.

Oil and Natural Gas Fields: a surface area above a particular pool of oil/gas reserves, where exploration or extraction occurs.

Oil and Natural Gas License Blocks: a surface area where a company or joint venture has been given the rights to explore or extract O&G resources.

Oil and Natural Gas Basins: a geological province, whose bounds are usually defined by the extent of certain rock formations above or below the surface.

Acknowledgments

We thank Kaiya Weatherby at Environmental Defense Fund for his help in quality-assuring the latest version of the database.

Database limitations

The database is based on public information which can be of variable spatial quality, update frequency, and richness of feature attributes depending on the data source. While we have made substantial improvement to the database in this version, we expect to further refine and QA/QC the database in future releases, especially as new data sources and updates to existing sources become available.

For comments, questions, or feedback, please contact:

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Supplement 1: OGIM v2.7 Database Schema

Table A: Attributes present in all layers

Attribute name	Data type	Allow nulls?	Description	Valid values and/or example values	Notes
CATEGORY	string	no	Category of O&G infrastructure to which the record belongs.	valid: 'CRUDE OIL REFINERIES'; 'EQUIPMENT AND COMPONENTS'; 'GATHERING AND PROCESSING'; 'INJECTION AND DISPOSAL'; 'LNG FACILITIES'; 'NATURAL GAS COMPRESSOR STATIONS'; 'NATURAL GAS FLARING DETECTIONS'; 'OFFSHORE PLATFORMS'; 'OIL AND NATURAL GAS BASINS'; 'OIL AND NATURAL GAS FIELDS'; 'OIL AND NATURAL GAS LICENSE BLOCKS'; 'OIL AND NATURAL GAS PIPELINES'; 'OIL AND NATURAL GAS WELLS'; 'PETROLEUM TERMINALS'; 'STATIONS - OTHER'; 'TANK BATTERIES'	Within a geopackage layer, all values for CATEGORY should be the same.
COUNTRY	string	no	Country in which the record resides. Where possible, country name matches the UN Member State list.	ex: 'GERMANY'; 'AFGHANISTAN, TURKMENISTAN'	Only LineString and Polygon features may fall in 2+ countries; in these cases, COUNTRY field contains a comma-separated list of these countries in alphabetical order.
OGIM_ID	integer	no	Unique identifier for each record in the geopackage. Values do not repeat across infrastructure categories.		
ON_OFFSHORE	string	no	Indicates whether the record lies onshore, offshore, or both.	valid: 'ONSHORE'; 'OFFSHORE'; 'ONSHORE, OFFSHORE'	Only LineString and Polygon features may fall both on and offshore, so only these geometries may have the value 'ONSHORE, OFFSHORE'.
REGION	string	yes	World region in which the record lies. When possible, region aligns with the IEA's Energy Region classifications.	valid: 'AFRICA', 'ASIA PACIFIC', 'CENTRAL AND SOUTH AMERICA', 'EURASIA', 'EUROPE', 'MIDDLE EAST', 'NORTH AMERICA', 'N/A'	It is possible for line and polygon features to have a value of N/A if they pass through multiple countries/regions.

Supplement 1: OGIM v2.7 Database Schema

SRC_DATE	string	no	Publication date of the record's original data source; YYYY-MM-DD format.	ex: '2024-06-01'	If a record's attributes were derived from 2+ different sources (see below), the most recent source date is listed in this field.
SRC_REF_ID	string	no	ID number(s) linking the record to its corresponding source in the "Data_Catalog" table.	ex: '22'; '89,92'	Records that list multiple SRC_REF_IDs separated by a comma (such as '89,92') indicate that different attributes of that record were obtained from different sources. For example, a record's location may have come from SRC_ID 89, but facility status came from SRC_ID 92.
STATE_PROV	string	yes	State or province in which the record resides.	ex: 'TEXAS'; 'ALBERTA'	LineString or Polygon features that intersect a very large number of states/provinces may simply list a value of 'N/A', to prevent excessively long attribute values.
geometry	geometry	no	Vertices of the feature's geometry. Formatted as well-known text (WKT) representations of the geometries.	ex: 'POINT (67.42377999999999 37.21161)'	

Supplement 1: OGIM v2.7 Database Schema

Table B: Attributes present in all well + infrastructure layers

Attribute name	Data type	Allow nulls?	Description	Valid values and/or example values	Notes
FAC_ID	string	yes	Unique ID used by the original source agency to identify the infrastructure asset.	ex: 'BGBR0230'; '126162'; '5609/10-01'	
FAC_NAME	string	yes	Name of the infrastructure asset.		
FAC_STATUS	string	yes	Operational status of the infrastructure asset, according to the original source.	ex: 'ACTIVE'; 'SUSPENDED'; 'TEMPORARILY CLOSED'	FAC_STATUS of "N/A" cannot be assumed to mean active or inactive.
FAC_TYPE	string	yes	Detailed information on type of facility.	ex: "EXPORT FACILITY"; "NGL FRACTIONATION FACILITY"	
INSTL_DATE *	string	yes	Date the facility or asset was installed; YYYY-MM-DD format.	ex: '1994-02-17'	Some data sources only included an installation year, or a month-year combination. These values appear with their month or date value filled with '01'. For example, if we only know a facility was installed in 2012, the INSTL_DATE would appear '2012-01-01'.
LATITUDE	float	no	Latitude of Point features in decimal degrees.	ex: 30.11438	
LONGITUDE	float	no	Longitude of Point features in decimal degrees.	ex: -93.29659	
OGIM_STATUS	string	yes	Standardized version of FAC_STATUS, created by the OGIM authors to "bin" statuses reported by the original source into categories.	valid: 'PERMITTING'; 'UNDER CONSTRUCTION'; 'OPERATIONAL'; 'PROPOSED'; 'DRILLING'; 'COMPLETED'; 'PRODUCING'; 'INACTIVE'; 'ABANDONED'; 'INJECTING'; 'STORAGE, MAINTENACE, OR OBSERVATION'; 'OTHER'	
OPERATOR	string	yes	Name of the asset's operator, according to the original source at time of publication.	ex: 'YSUR ENERGÍA ARGENTINA S.R.L.'; 'PETROBRAS'; 'DCP MIDSTREAM, LP'	No modifications have been made to standardize operator names or associate subsidiaries with parent companies.

* = attribute not present for wells

Supplement 1: OGIM v2.7 Database Schema

Table C: Attributes present in wells layer only

Attribute name	Data type	Allow nulls?	Description	Valid values and/or example values	Notes
COMP_DATE	string	yes	Date that well construction was completed; YYYY-MM-DD format.	ex: '2019-12-13'	
DRILL_TYPE	string	yes	Drilling direction of the well.	ex: 'HORIZONTAL'; 'VERTICAL'; 'DIRECTIONAL'; 'CONVENTIONAL'	Conventional' indicates a vertical well; 'Unconventional' indicates a horizontal or hydraulic fracturing well.
SPUD_DATE	string	yes	Date that well was first spudded (i.e., drilling began); YYYY-MM-DD format.	ex: '2019-04-11'	

Table D: Attributes present in pipelines only

Attribute name	Data type	Allow nulls?	Description	Valid values and/or example values	Notes
PIPE_DIAMETER_MM	float	yes	Pipeline diameter in millimeters.	ex: 88, 114	
PIPE_LENGTH_KM	float	no	Length of pipeline feature in kilometers.	ex: 4.45; 90.4; 1130	Pipeline length is calculated for each feature in GIS by the OGIM authors, even if the original data source reports a length.
PIPE_MATERIAL	float	yes	Material that pipeline is made of.	ex: 'STEEL'; 'POLYETHYLENE'; 'CARBON STEEL 5L GRADE X 65'; 'ASTM A-106 GR.B'	

Table E: Attributes present in basins, fields, and license blocks layers only

Attribute name	Data type	Allow nulls?	Description	Valid values and/or example values	Notes
NAME	string	yes	The name of the basin, field, or license block.	ex: 'PERMIAN'	
AREA_KM2	float	no	Area of polygon in sq. kilometers.	ex: 37200; 186000	Area is calculated for each feature in GIS by the OGIM authors, even if the original data source reports an area.
RESERVOIR_TYPE	string	yes	Hydrocarbon(s) produced by the reservoir, OR the phase of production the reservoir is in.	ex: 'OIL'; 'OIL AND GAS'; 'CONDENSATE'; 'EXPLORATION AND EXPLOITATION'; 'COALBED METHANE'	

Table F: Attributes present in flaring detections only

Attribute name	Data type	Allow nulls?	Description	Valid values and/or example values	Notes
AVERAGE_FLARE_TEMP_K	integer	no	Average flare temperature in Kelvin.	ex: 1020, 2119	
DAYS_CLEAR_OBSERVATIONS	integer	no	Number of clear days for which flares were detected.	ex: 123, 381	
FLARE_YEAR	integer	no	Year in which detections occurred.	valid: 2023	
GAS_FLARED_MMCf	float	no	Estimated volume of gas flared in million cubic feet per year.		
SEGMENT_TYPE	string	yes	Oil and gas industry segment to which the flaring detection belongs.	valid: 'GAS DOWNSTREAM'; 'OIL DOWNSTREAM'; 'UPSTREAM'	

Table H: Additional attributes in infrastructure layers

Attribute name	Data type	Allow nulls?	Description	In which layer(s) is attribute present?
GAS_CAPACITY_MMCF D	float	yes	Facility capacity for natural gas, in million cubic ft. per day.	CS, GP, ID, LNG, PL, TM
GAS_THROUGHPUT_M MCFD	float	yes	Facility throughput for natural gas, in million cubic ft. per day.	CS, GP, ID, LNG, PL, TM
LIQ_CAPACITY_BPD	float	yes	Facility capacity for O&G liquids, in barrels per day.	CS, GP, ID, LNG, PL, R, TM
LIQ_THROUGHPUT_BP D	float	yes	Facility throughput for O&G liquids, in barrels per day.	CS, GP, ID, LNG, PF, PL, R, TM
NUM_STORAGE_TANKS	integer	yes	Number of storage tanks at the facility.	CS, GP, ID, LNG, PF, R, TM
NUM_COMPR_UNITS	integer	yes	Number of compressor units present at facility.	CS, GP, TM
SITE_HP	float	yes	Horsepower of the facility.	CS, GP
COMMODITY	string	yes	Hydrocarbon(s) contained in the infrastructure.	PL, SO, TM

Key for Table H:

CS = Compressor Stations

GP = Gathering and Processing

ID = Injection and Disposal

LNG = Liquefied Natural Gas Facilities

PL = Pipelines

R = Refineries

SO = Stations (Other)

TM = Petroleum Terminals

Table Z: Attributes present in the Data Catalog layer

Attribute name	Data type	Allow nulls?	Description	Valid values and/or example values
SRC_ID	integer	no	Identification number unique to the particular data source.	
SRC_URL	string	no	URL to the webpage that describes the data and offers it for download.	
SRC_NAME	string	no	Source agency or name (written out fully).	ex: 'New Zealand Petroleum & Minerals'; 'Ministero della transizione ecologic'
SRC_ALIAS	string	yes	Optional short-name for source; may be an abbreviation of the source agency, or the name/username of the original data author.	ex: 'NZPAM'; '@jesse_hamlin'; 'USGS'
SRC_TYPE	string	no	General categorization of the source type, as determined by OGIM team.	valid: 'Academia', 'ArcGIS Online', 'Company', 'Data Vendor', 'Government', 'NGO', 'OGInfra.com'
SRC_YEAR	integer	no	Year that source was published.	valid: integers 1900 - 2024 inclusive
SRC_MNTH	integer	no	Month that source was published.	valid: integers 1 - 12 inclusive
PUB_PRIV	string	no	Public availability of data.	valid: 'Public'; 'Proprietary'
PROD_DATA	boolean	no	Does the original data source report O&G production values?	valid: 0,1
UPDATE_FREQ	string	no	Frequency with which the source is usually updated.	valid: 'Annually', 'Quarterly'; 'Monthly', 'Weekly'; 'Daily'; 'Irregularly'; 'One-Time'; 'Uncertain'
LASTVISIT	string	no	Date on which an EDF analyst last visited the source URL; YYYY-MM-DD format.	ex: '2024-12-13'
REGION	string	no	Region in which data lies.	valid: 'Africa'; 'Asia'; 'Central Asia'; 'Europe'; 'Global'; 'Middle East'; 'North America'; 'South America'; 'Oceania'; 'Various'
COUNTRY	string	yes	If applicable, country in which data lies.	ex: 'Argentina'; 'Various'
STATE_PROV	string	yes	If applicable, state or province in which data lies.	ex: 'Various'; 'Sub-national'; 'Kentucky'; 'Alberta'
NOTES	string	yes	Internal notes about the source or its contents.	
HAS_WELLS	boolean	no	Does this data source contain any records that are included in the Oil and Natural Gas Wells layer?	valid: 0, 1
HAS_PLATFORMS	boolean	no	Does this data source contain any records that are included in the Offshore Platforms layer?	valid: 0, 1

Supplement 2: OGIM v2.7 Data Catalog Schema

HAS_COMPRESSORS	boolean	no	Does this data source contain any records that are included in the Natural Gas Compressor Stations layer?	valid: 0, 1
HAS_PROCESSING	boolean	no	Does this data source contain any records that are included in the Gathering and Processing layer?	valid: 0, 1
HAS_REFINERIES	boolean	no	Does this data source contain any records that are included in the Crude Oil Refineries layer?	valid: 0, 1
HAS_TERMINALS	boolean	no	Does this data source contain any records that are included in the Petroleum Terminals layer?	valid: 0, 1
HAS_LNG	boolean	no	Does this data source contain any records that are included in the LNG Facilities layer?	valid: 0, 1
HAS_INJ_DISP	boolean	no	Does this data source contain any records that are included in the Injection and Disposal layer?	valid: 0, 1
HAS_TANKS	boolean	no	Does this data source contain any records that are included in the Tank Battery layer?	valid: 0, 1
HAS_EQUIP_COMP	boolean	no	Does this data source contain any records that are included in the Equipment and Components layer?	valid: 0, 1
HAS_OTHER	boolean	no	Does this data source contain any records that are included in the Stations - Other layer?	valid: 0, 1
HAS_PIPELINES	boolean	no	Does this data source contain any records that are included in the Oil and Natural Gas Pipelines layer?	valid: 0, 1
HAS_BASINS	boolean	no	Does this data source contain any records that are included in the Oil and Natural Gas Basins layer?	valid: 0, 1
HAS_FIELDS	boolean	no	Does this data source contain any records that are included in the Oil and Natural Gas Fields layer?	valid: 0, 1
HAS_BLOCKS	boolean	no	Does this data source contain any records that are included in the Oil and Natural Gas License Blocks layer?	valid: 0, 1
HAS_PRODUCTION	boolean	no	Does this data source contain any records that are included in the Oil and Natural Gas Production layer?	valid: 0, 1
HAS_FLARES	boolean	no	Does this data source contain any records that are included in the Natural Gas Flaring Detections layer?	valid: 0, 1